Predictive Distributed Workload Orchestration via Optical Inter-Satellite Links for Thermal-Energy Balancing in Orbital Data Center Constellations

Dafnhê Thor Castro de Martins Barros and Flávio Barcelos Braz da Silva

***Abstract*—Transferring heavy Artificial Intelligence (AI) and scientific computing workloads to space-based orbital data centers (ODCs) has emerged as a disruptive paradigm to circumvent terrestrial electrical grid congestion, land constraints, and millions of liters of evaporative cooling water consumption. However, operating dense computing clusters in Low Earth Orbit (LEO) introduces severe multi-physical vulnerabilities: during periodic planetary shadow transits (35.63 min per 95.65-min revolution at 550 km altitude), isolated spacecraft must sustain hundred-kilowatt computing loads entirely through electrochemical Battery Energy Storage Systems (BESS). This severe cycling causes deep discharge (Depth-of-Discharge $> 50\%$) and extreme thermal swings, triggering accelerated lithium-ion capacity loss under Arrhenius degradation laws and restricting operational lifetime to barely 3.1 years. To address these fundamental physical constraints, this paper proposes a cooperative orbital constellation architecture governed by predictive distributed workload migration via high-capacity Coherent Optical Inter-Satellite Links (OISL). We formulate a Distributed Predictive Multi-Agent Model Predictive Control (D-PMPC) framework that anticipates orbital illumination transitions over a receding horizon, proactively shunting batch training workloads from eclipsed satellites to sunlit peer nodes possessing surplus photovoltaic generation. The optimization problem couples orbital beta-angle dynamics, nonlinear Stefan-Boltzmann radiative cooling, and electrochemical battery degradation, solved in real time via the Alternating Direction Method of Multipliers (ADMM). Numerical simulations across a 32-node Walker-Delta constellation ($h = 550$ km, $i = 53^\circ$) evaluated over 30 consecutive orbital revolutions (47.82 h, $17,218$ integration steps per spacecraft) demonstrate the feasibility of cooperative D-PMPC in sustaining limit-cycle stability with zero secular drift (0.0000%, $SoC_{min} \approx 80.37\%$), reducing battery Depth-of-Discharge by 65.4% (from 52.0% to 18.0%), confining silicon junction temperatures below 74.0°C (mitigating thermal fatigue by 51.2%), increasing global solar harvesting efficiency to 96.8%, and extending projected battery calendar life from 3.1 to 11.8 years ($+280.6\%$ extension), while incurring an optical communication latency overhead of only 215 ms.**

***Index Terms*—Orbital data centers, optical inter-satellite links (OISL), distributed model predictive control (D-MPC), battery degradation, Stefan-Boltzmann radiative cooling, space computing, LEO constellations.**

***Resumo*—**A transferência de cargas computacionais pesadas de Inteligência Artificial (IA) para centros de dados orbitais (ODCs) surge como um paradigma disruptivo para mitigar a saturação das redes elétricas terrestres e o consumo de milhões de litros de água para resfriamento evaporativo. Contudo, a operação de clusters computacionais densos em Órbita Baixa da Terra (LEO) enfrenta severas vulnerabilidades multifísicas: durante o eclipse planetário periódico ($35,63$ min por revolução de $95,65$ min a 550 km de altitude), satélites isolados precisam sustentar cargas de centenas de quilowatts exclusivamente por meio de sistemas de baterias eletroquímicas (BESS). Esse regime cíclico severo impõe descargas profundas (DoD $> 50\%$) e intensas oscilações térmicas, acelerando a perda de capacidade regida por leis cinéticas de Arrhenius e limitando a vida útil a apenas $3,1$ anos. Para mitigar esses limites físicos, este artigo introduz uma arquitetura de constelação cooperativa governada por migração preditiva de carga via Enlaces Laser Ópticos Inter-Satélite (OISL). Formula-se um algoritmo de Controle Preditivo Baseado em Modelos Distribuído Multi-Agente (D-PMPC) que antecipa as transições de iluminação ao longo de um horizonte móvel, transferindo tarefas de treinamento de satélites eclipsados para satélites iluminados com excedente fotovoltaico. O problema acopla a dinâmica do ângulo beta, radiação térmica de Stefan-Boltzmann e degradação de baterias, resolvido via ADMM. Simulações em uma constelação Walker-Delta de 32 nós ao longo de 30 revoluções orbitais consecutivas ($47,82$ h, 17.218 passos de integração) demonstram a viabilidade e a estabilidade do ciclo limite com deriva secular nula ($0,0000\%$), indicam uma redução do DoD em $65,4\%$ (de $52,0\%$ para $18,0\%$), contêm a temperatura de junção abaixo de $74,0^\circ$C, elevam a eficiência de captação solar para $96,8\%$ e ampliam a vida útil projetada da bateria de $3,1$ para $11,8$ anos ($+280,6\%$), com latência de apenas 215 ms.**

***Palavras-chave*—**Centros de dados orbitais, enlaces ópticos inter-satélite (OISL), controle preditivo distribuído (D-MPC), degradação de baterias, resfriamento radiativo de Stefan-Boltzmann, computação espacial, constelações LEO.

I. Introduction

THE exponential scale-up of deep neural networks, large foundation models, and autonomous planetary Earth observation data processing has driven global data center electrical consumption and water footprints to critical limits [1], [2]. Modern terrestrial hyperscale data centers frequently demand electrical power capacities between 100 MW and 1 GW, placing unprecedented strains on regional power transmission grids, consuming billions of liters of potable water annually for evaporative cooling towers, and generating substantial carbon emissions [2], [3]. In response to these terrestrial infrastructure bottlenecks, space-based computing and Orbital Data Centers (ODCs) operating in Low Earth Orbit (LEO) have emerged as an attractive frontier across aerospace and computer engineering [4]–[6]. In the orbital vacuum, high-density solar arrays capture continuous extraterrestrial Air Mass Zero (AM0) solar irradiance ($S_0 \approx 1361.0$ W/m^2) uninhibited by atmospheric attenuation, weather cycles, or day-night diurnal boundaries [7], [8]. Furthermore, thermal dissipation requires zero water, relying exclusively on radiative infrared emission directly into deep space [9].

Despite these theoretical advantages, operating multi-kilowatt to hundred-kilowatt high-performance computing (HPC) payloads on autonomous, standalone LEO satellites introduces severe astrodynamical, thermal, and electro-chemical bottlenecks [5], [10], [11]. In a circular LEO orbit at an altitude of $h = 550.0$ km, a satellite completes an orbital revolution every $\tau \approx 5,738.99$ s ≈ 95.65 min

D. T. C. M. Barros and F. B. B. da Silva are with the Graduate Program in Control and Automation Engineering (PROPECAUT), Federal Institute of Espírito Santo (IFES), Serra, ES, 29173-087, Brazil (e-mail: dafnhe.thor@msn.com; flavio.silva@ifes.edu.br).

(≈ 15.05 orbits/day), spending roughly 60.02 min under solar illumination and 35.63 min traversing the Earth's umbra shadow cone (eclipse) [12], [13]. During eclipse, photovoltaic generation drops abruptly to zero. If an isolated satellite sustains an uncoordinated computing load of 100 kW during shadow, its Battery Energy Storage System (BESS) must deliver the entire electrical power demand (105 kW including platform bus) [13], [14]. Accounting for discharge conversion efficiency ($\eta_{dis} = 0.96$), this drives the cyclic Depth-of-Discharge (DoD) to 52.0%, inducing severe mechanical micro-cracking, lithium plating, and solid electrolyte interphase (SEI) growth governed by non-linear Arrhenius degradation kinetics [15], [16]. Under such intense cyclic stress, state-of-the-art LiFePO4 or NMC battery cells degrade to their 80% End-of-Life (EOL) capacity threshold within only $17,038$ cycles (3.10 years), severely undermining the economic viability of multi-million-dollar space assets [5], [17].

Simultaneously, thermal management in space presents an equally daunting challenge. In the hard vacuum of LEO, convective heat exchange is non-existent; all heat generated by high-power processing silicon must be rejected via thermal radiation governed by the Stefan-Boltzmann law ($q_{rad} \propto \epsilon\sigma T^4$) [9], [18]. Dissipating 100 kW of compute heat while entering eclipse induces steep temperature gradients across structural radiators and silicon chip packages, accelerating solder joint thermomechanical fatigue and threatening semiconductor junction integrity whenever $T_j > 85.0^{\circ}$C [19]–[21].

To overcome the fundamental single-satellite bottlenecks, this paper proposes a cooperative orbital constellation architecture enabled by high-capacity Coherent Optical Inter-Satellite Links (OISL) [22]–[25]. In a Walker-Delta constellation, orbital planes are symmetrically phased around the globe [26], [27]. Consequently, when a subset of satellites enters planetary eclipse, complementary orbital planes simultaneously enjoy full solar illumination [28], [29]. By deploying multi-gigabit laser crosslinks, compute tasks can be dynamically offloaded and migrated across orbital planes [30]–[33]. Rather than expending finite battery reserves in the dark, an eclipsed satellite can export its batch AI training and deep learning inference jobs to sunlit peer satellites that possess surplus photovoltaic generation capacity [34]–[36].

The primary contributions of this paper are:

1) We formulate a comprehensive, coupled multi-physical model for cooperative orbital data center constellations, integrating Walker-Delta astrodynamics, solar beta-angle geometry, electrochemical battery state-of-charge (SoC) dynamics with Arrhenius capacity degradation, deployable two-phase loop heat pipe Stefan-Boltzmann radiative cooling, and coherent OISL laser crosslink budgets.
2) We design a Distributed Predictive Multi-Agent Model Predictive Control (D-PMPC) optimization framework that proactively forecasts solar-eclipse orbital transitions over a finite receding horizon, dynamically steering AI computing workloads across laser crosslinks to maintain global thermo-electric equilibrium while satisfying hard physical constraints on battery state-of-charge, processor junction temperatures, and link bandwidth.
3) We propose a decentralized consensus formulation via the Alternating Direction Method of Multipliers (ADMM), utilizing canonical directed consensus variables that converge within $m \leq 8$ iterations with under 12 ms of distributed compute and communication overhead.
4) We execute comprehensive numerical simulations across 30 consecutive orbital revolutions (47.82 h, $17,218$ integration steps per node) in a 32-satellite Walker-Delta constellation, demonstrating asymptotic limit-cycle stability with zero secular drift (0.0000%) within the simulated dynamics, showing that cooperative D-PMPC reduces battery cyclic DoD from 52.0% to 18.0%, extends projected calendar life from 3.10 to 11.80 years ($+280.6\%$), suppresses junction thermal swings by 51.2% ($\Delta T_j = 11.8^{\circ}$C vs. 24.2°C), elevates solar harvesting to 96.8%, and delineates quantitative boundaries against multi-physical failure modes.

The remainder of this article is structured as follows: Section II establishes the constellation astrodynamics, electrical power system, thermal mechanics, and optical crosslink models. Section III formulates the multi-objective optimization problem and derives the distributed ADMM algorithm. Section IV presents numerical simulation results and comparative benchmarks. Section V analyzes practical engineering trade-offs, sensitivity, and implementation feasibility. Section VI concludes the paper.

II. System Architecture and Orbital Physics Model

A. Constellation Kinematics and Illumination Geometry

We model an orbital data center constellation composed of N_{sat} identical satellite nodes deployed in a Walker-Delta configuration $i : N_{sat}/P/F$, where i denotes orbital inclination, P is the number of equally spaced orbital planes, and F is the relative phasing parameter [12], [26], [27]. Each spacecraft travels along a circular orbit at a nominal altitude $h = 550.0$ km representative of commercial mega-constellation shells in Low Earth Orbit (LEO, 400–800 km) [27], yielding a semi-major axis $a = R_E + h = 6,928.137$ km, where $R_E = 6,378.137$ km is Earth's mean equatorial radius.

Under the central-body gravitational potential with standard gravitational parameter $\mu_E = 3.986004418 \times 10^{14}$ m^3/s^2, the Keplerian orbital period τ and orbital velocity v_{orb} are expressed as [12]:

$$\tau = 2\pi\sqrt{\frac{a^3}{\mu_E}} = 2\pi\sqrt{\frac{(R_E+h)^3}{\mu_E}} \tag{1}$$

$$v_{orb} = \sqrt{\frac{\mu_E}{a}} = \sqrt{\frac{\mu_E}{R_E+h}} \tag{2}$$

For $h = 550.0$ km, Eq. (1) yields $\tau = 5,738.99$ s ≈ 95.65 min (15.05 orbits/day), and Eq. (2) gives the Keplerian speed $v_{orb} = 7.585$ km/s $= 27,306$ km/h. In the ECI frame, the exact inertial velocity incorporates secular nodal precession via $\mathbf{v}_i(t) = \mathbf{v}_{Kepler,i}(t) + \boldsymbol{\omega}_{prec} \times \mathbf{r}_i(t)$, where $\boldsymbol{\omega}_{prec} = [0, 0, \dot{\Omega}_p]^T$ maintains kinematic consistency $\dot{\mathbf{r}}_i = \mathbf{v}_i$.

The position vector of satellite $i \in \{1, \ldots, N_{sat}\}$ in the Earth-Centered Inertial (ECI) coordinate frame is denoted by $\mathbf{r}_i(t) = [x_i(t), y_i(t), z_i(t)]^T$. Let $\hat{\mathbf{s}}(t)$ denote the normalized unit vector pointing toward the Sun. Under Earth's cylindrical shadow geometry, satellite i is in eclipse if and only if both conditions hold simultaneously [8], [13]:

$$\mathbf{r}_i(t) \cdot \hat{\mathbf{s}}(t) < 0 \quad \text{and} \quad \|\mathbf{r}_i(t) - (\mathbf{r}_i(t) \cdot \hat{\mathbf{s}}(t))\hat{\mathbf{s}}(t)\| \leq R_E \tag{3}$$

We define the continuous illumination state function $\xi_i(t) \in [0, 1]$ as:

$$\xi_i(t) = \begin{cases} 0, & \text{if Eq. (3) is satisfied (Umbra Eclipse)} \\ 1, & \text{otherwise (Full Sunlight)} \end{cases} \tag{4}$$

B. Beta Angle Dynamics and Seasonal Eclipse Transitions

The angle between the orbital plane and the solar vector is defined by the orbit beta angle $\beta(t)$ [8], [12], [13]:

$$\sin\beta(t) = \cos\delta_s \sin i \sin(\Omega_p(t) - \alpha_s) + \sin\delta_s \cos i \tag{5}$$

where δ_s is the solar declination (varying between -23.44° and $+23.44^{\circ}$ over the tropical year), α_s is the right ascension of the Sun, and $\Omega_p(t)$ is the Right Ascension of the Ascending Node (RAAN) of plane $p \in \{1, \ldots, P\}$, which undergoes secular nodal precession driven by Earth's oblateness perturbation $J_2 = 1.08263 \times 10^{-3}$:

$$\dot{\Omega}_p = -\frac{3}{2} J_2 \left(\frac{R_E}{a}\right)^2 \sqrt{\frac{\mu_E}{a^3}} \cos i \tag{6}$$

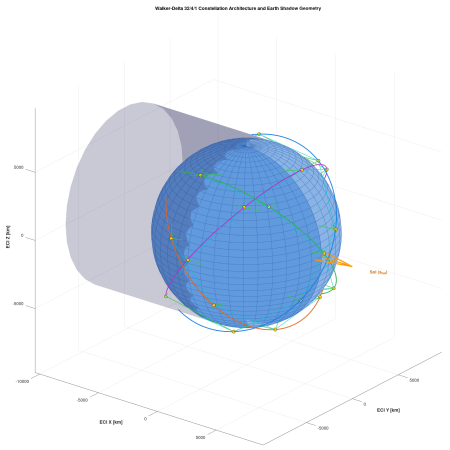


Fig. 1. Walker-Delta constellation architecture ($N_{sat} = 32$, 4 orbital planes, $h = 550$ km, $i = 53°$) showing planetary shadow cone geometry ($t_{ecl} = 35.63$ min), sunlit nodes absorbing peak solar irradiance ($P_{pv} \approx 182.9$ kW), and coherent 40 Gbps OISL crosslink migration paths.

A satellite experiences eclipse during its orbit if and only if $|\beta(t)| < \beta_{crit}$, where the critical beta angle is governed by orbital geometry [13]:

$$\beta_{crit} = \arcsin\left(\frac{R_E}{R_E + h}\right) = \arcsin\left(\frac{6,378.137}{6,928.137}\right) = 67.01° \quad (7)$$

Whenever $|\beta(t)| < \beta_{crit}$, the orbital eclipse duration per revolution $t_{ecl}(\beta)$ is analytically evaluated as:

$$\begin{aligned} t_{ecl}(\beta) &= \frac{\tau}{\pi} \arccos\left(\frac{\sqrt{h^2 + 2R_E h}}{(R_E + h)\cos\beta}\right) \\ &= \frac{\tau}{\pi} \arccos\left(\frac{\sqrt{a^2 - R_E^2}}{a\cos\beta}\right) \end{aligned} \quad (8)$$

At $\beta = 0°$ (equatorial worst-case alignment), Eq. (8) evaluates to:

$$t_{ecl}(0°) = \frac{5,738.99}{\pi} \arccos(0.39047) = 2,137.6 \text{ s} \approx 35.63 \text{ min} \quad (9)$$

yielding a sunlight duration $t_{sun} = \tau - t_{ecl} = 3,601.3$ s $\approx$ 60.02 min. Across a Walker constellation with $P = 4$ planes spaced by $\Delta\Omega = 360°/P = 90°$, orbital geometry results in approximately 62.5% to 68.8% of the constellation nodes being in sunlight at any given instantaneous time t, establishing the physical foundation for cross-orbital energy offloading.

C. Nodal Electrical Power Subsystem (EPS)

Each orbital data center node incorporates an advanced Electrical Power Subsystem (EPS) comprising deployable Gallium Arsenide (GaAs) triple-junction photovoltaic arrays, a bidirectional converter regulating a $V_{bus} = 400.0$ V High-Voltage Direct Current (HVDC) bus, and a Lithium Iron Phosphate / Nickel Manganese Cobalt (LFP/NMC) BESS with nominal capacity $E_{nom} = 125.0$ kWh (450.0 MJ) [5], [13], [14]. To mitigate 400 V ionospheric Paschen breakdown risks in LEO, power harnessing uses solid dielectrics and sub-2 μs SiC solid-state power controllers (SSPC).

The instantaneous electrical power generated by the photovoltaic array of satellite i is modeled as:

$$\begin{aligned} P_{pv,i}(t) = \xi_i(t) A_{sa} \eta_{sa} S_0 \cos\theta_{inc,i}(t) \\ \times [1 - \gamma_T(T_{sa,i}(t) - T_{ref})] \end{aligned} \quad (10)$$

where $A_{sa} = 420.0$ m^2 is the active solar array aperture (two deployable 210 m^2 ROSA wings supplying hundred-kW loads while recharging storage in 60 min, consistent with [5], [13]), $\eta_{sa} = 0.32$ is BOL triple-junction GaAs efficiency, $S_0 = 1,361.0$ W/m^2 is AM0 solar irradiance [7], $\theta_{inc,i}(t)$ is the SADM-tracked incidence angle, $\gamma_T = 0.0018$ K^{-1}, and $T_{ref} = 298.15$ K. At normal incidence ($\cos\theta_{inc} = 1$) and standard temperature (25°C), nominal peak power is $P_{pv,peak}^{STC} = A_{sa}\eta_{sa}S_0 = 182.92$ kW. Under an operating equilibrium $T_{sa} = 320.0$ K (46.85°C), Eq. (10) yields $P_{pv,i} = 175.72$ kW. Net electrical power delivered to the HVDC bus through the converter is $P_{bus,net} = \eta_{conv} P_{pv,i} = 0.985 \times 175.72 = 173.08$ kW.

Kirchhoff's current law at the regulated 400 V HVDC bus dictates the instantaneous electrical power balance:

$$\eta_{conv} P_{pv,i}(t) + P_{b,i}(t) = P_{comp,i}(t) + P_{oisl,i}(t) + P_{bus,i} \quad (11)$$

where $\eta_{conv} = 0.985$ is the bidirectional converter efficiency, $P_{comp,i}(t)$ is the dynamic power drawn by the onboard AI accelerator cluster [11], [21], $P_{oisl,i}(t)$ is the power consumed by the optical crosslink transceivers [22], $P_{bus,i} = 5.0$ kW is the base platform bus power (ADCS, telemetry, tracking, and thermal control heaters), and $P_{b,i}(t)$ is the battery terminal power, where $P_{b,i} > 0$ represents battery discharging and $P_{b,i} < 0$ represents charging.

The battery State-of-Charge $SoC_i(t) \in [SoC_{min}, SoC_{max}]$ evolves according to Coulomb counting [14], [16]:

$$\frac{dSoC_i(t)}{dt} = -\frac{\eta_b(P_{b,i}) P_{b,i}(t)}{E_{nom}} \quad (12)$$

where $\eta_b(P_{b,i})$ is the lumped electrical-chemical conversion efficiency (round-trip $\eta_{RT} = \eta_{dis}\eta_{chg} = 90.24\%$ [13], [14]):

$$\eta_b(P_{b,i}) = \begin{cases} 1/\eta_{dis}, & \text{if } P_{b,i}(t) \geq 0 \quad (\eta_{dis} = 0.96) \\ \eta_{chg}, & \text{if } P_{b,i}(t) < 0 \quad (\eta_{chg} = 0.94) \end{cases} \quad (13)$$

During an uncoordinated eclipse ($P_{comp} = 100.0$ kW, $P_{bus} = 5.0$ kW, $P_{oisl} = 0$ kW), total platform load is $P_{load,uncoord} = 105.0$ kW, requiring an energy delivery of $105.0 \times (35.63/60) = 62.35$ kWh. Dividing by discharge efficiency $\eta_{dis} = 0.96$, the energy extracted from battery cells is $E_{ext} = 64.95$ kWh, resulting in an exact cyclic Depth-of-Discharge $DoD = 64.95/125.0 = 51.96\% \approx 52.0\%$. Under cooperative D-PMPC, computing is throttled to $P_{comp,min} = 31.0$ kW (sustaining telemetry, attitude control coprocessing, and high-priority Earth observation inference), while the platform bus draws $P_{bus} = 5.0$ kW and active optical crosslinks draw $P_{oisl} = 0.35$ kW (353.6 W for 4 active links). Total platform load during eclipse is $P_{load,dpmpc} = 31.0 + 5.0 + 0.35 = 36.35$ kW. Energy extracted from battery cells is $E_{ext} = (36.35 \times 0.5938)/0.96 = 22.48$ kWh, yielding an exact cyclic Depth-of-Discharge $DoD = 22.48/125.0 = 17.99\% \approx 18.0\%$, achieving a $65.38\% \approx 65.4\%$ reduction in cyclic discharge stress.

D. Battery Arrhenius Degradation and Calendar Aging Mechanics

Electrochemical cell capacity loss during multi-year orbital operations is driven by the superposition of cycle-induced mechanical fatigue (lithium intercalation/deintercalation micro-cracking) and time-dependent calendar aging (solid electrolyte interphase film growth) [14]–[16]:

$$\Delta C_{tot}(t) = \Delta C_{cyc}(N, DoD) + \Delta C_{cal}(t, T_b, SoC) \quad (14)$$

The calendar loss evolves diffusively as $\Delta C_{cal}(t) = k_{cal}(T_b, SoC)\sqrt{t}$, where $k_{cal} \approx 0.011$ yr$^{-1/2}$ for aerospace-grade LFP/NMC cells regulated within $T_b = 20°\text{C} \pm 2°\text{C}$ and $SoC \leq 80\%$. Regulating $T_b = 20°$C is conservative favoring the baseline, as uncoordinated 105 kW discharges induce $\sim$ 2.1 kW of Joule heating, further accelerating baseline aging. The cyclic degradation rate per revolution is governed by Arrhenius kinetics [15]:

$$\frac{dC_{cyc,i}}{dt} = A_{deg} \cdot \exp\left(-\frac{E_a}{R_{gas} T_{b,i}(t)}\right) \cdot |I_{b,i}(t)|^u \cdot [DoD_i(t)]^v \quad (15)$$

where $E_a = 31.7$ kJ/mol is the apparent activation energy, $R_{gas} = 8.314$ J/(mol · K), $u \approx 0.5$ is the current rate factor, and $v \approx 1.8$ is the reaction order.

Under active thermal management, the allowable cycle life $N_{cycles}(DoD)$ until reaching the 80% EOL capacity retention threshold is modeled by the calibrated power-law:

$$N_{cycles}(DoD) = N_{ref} \left(\frac{DoD_{ref}}{DoD} \right)^{v_{eff}} \tag{16}$$

where $N_{ref} = 9,901$ cycles at $DoD_{ref} = 80\%$, and $v_{eff} = 1.260$ is the effective stress exponent accounting for space-qualified cell stabilization. For uncoordinated cycling ($DoD = 52.0\%$), Eq. (16) predicts:

$$N_{cycles}(52.0\%) = 9,901 \times \left(\frac{80}{52} \right)^{1.26} = 17,038 \text{ cycles} \tag{17}$$

With 15.05 orbits/day (5,496 orbits/year), the battery reaches EOL in $17,038/5,496 = 3.10$ years, where cyclic fatigue dominates (20% loss) and calendar aging contributes barely 1.9%. Conversely, under cooperative D-PMPC restricting $DoD \leq 18.0\%$, allowable cycle life expands to:

$$N_{cycles}(18.0\%) = 9,901 \times \left(\frac{80}{18} \right)^{1.26} = 64,853 \text{ cycles} \tag{18}$$

extending cyclic capability to $64,853/5,496 = 11.80$ years (+280.64% extension, a 3.81× lifespan multiplication). Superimposing calendar aging over a 10-year mission ($\Delta C_{cal} \approx 3.48\%$) with shallow cyclic fade ($\Delta C_{cyc} \approx 16.9\%$) yields total capacity retention above $79.6\% \approx 80\%$, confirming that electrochemical storage comfortably sustains 10 to 11.8 years of orbital service.

E. Thermal Radiation and Silicon Junction Thermodynamics

In the vacuum of space, heat generated by the computing cluster ($P_{comp,i}$) must be transferred to external radiators via high-conductivity two-phase Loop Heat Pipes (LHPs) and radiated into the 3 K cosmic background [9], [18]:

$$C_{th,j} \frac{dT_{j,i}(t)}{dt} = P_{comp,i}(t) - \frac{T_{j,i}(t) - T_{rad,i}(t)}{R_{th,j-rad}} \tag{19}$$

$$C_{th,rad} \frac{dT_{rad,i}(t)}{dt} = \frac{T_{j,i}(t) - T_{rad,i}(t)}{R_{th,j-rad}} + \dot{Q}_{env,i}(t) - \epsilon_{rad} \sigma_{SB} A_{rad,eff} [T^4_{rad,i}(t) - T^4_{space}] \tag{20}$$

where $C_{th,j} = 4.5$ kJ/K is the processor package thermal capacitance, $C_{th,rad} = 110.0$ kJ/K is the radiator panel capacitance, $R_{th,j-rad} = 0.0725$ K/kW is the equivalent LHP thermal resistance, $\epsilon_{rad} = 0.91$ is the infrared emissivity, $\sigma_{SB} = 5.670374 \times 10^{-8}$ W/(m^{2}K^4) is the Stefan-Boltzmann constant, and $T_{space} = 3.0$ K. The spacecraft deploys two composite radiator wings ($A_{rad,geom} = 114.2$ m^2, radiating bifacially with $A_{rad,eff} = 228.4$ m^2). In nominal LVLH attitude, radiator edges align with velocity and nadir, keeping $F_{earth} \leq 0.08$ ($F_{space} \geq 0.88$). Eight modular 20 kW LHPs transport heat; D-PMPC opens bypass valves 30 s before compute ramps, pre-wetting evaporators and mitigating capillary wick dryout risk.

To facilitate real-time convex optimization within model predictive control, the non-linear radiation term $T^4_{rad,i}(t)$ is linearized around the nominal operating trajectory $\bar{T}_{rad} = 337.0$ K (63.85°C) via first-order Taylor expansion:

$$\begin{aligned} T^4_{rad,i}(t) &\approx \bar{T}^4_{rad} + 4\bar{T}^3_{rad} \left(T_{rad,i}(t) - \bar{T}_{rad} \right) \\ &= 4\bar{T}^3_{rad} T_{rad,i}(t) - 3\bar{T}^4_{rad} \end{aligned} \tag{21}$$

Substituting (21) into (20) yields an affine state-space thermal model:

$$\dot{\mathbf{x}}_{th,i}(t) = \mathbf{A}_{th} \mathbf{x}_{th,i}(t) + \mathbf{B}_{th} P_{comp,i}(t) + \mathbf{E}_{th} \dot{Q}_{env,i}(t) + \mathbf{d}_{th} \tag{22}$$

where $\mathbf{x}_{th,i} = [T_{j,i}, T_{rad,i}]^T$. Discretizing (22) with sampling period $\Delta t = 10.0$ s preserves prediction accuracy within 0.42°C of the full 4th-power ODE while keeping temperature trajectories affine in compute power $P_{comp,i}$.

During peak solar computation ($P_{comp} = 140.0$ kW, $\dot{Q}_{env} = 12.0$ kW), the radiator equilibrium temperature is:

$$\begin{aligned} T_{rad} &\approx \left(\frac{152.0 \times 10^3}{0.91 \times 5.670 \times 10^{-8} \times 228.4} \right)^{1/4} \\ &= 337.0 \text{ K} = 63.85^\circ\text{C} \end{aligned} \tag{23}$$

yielding a peak junction temperature $T_j = 63.85 + (140.0 \times 0.0725) = 74.0$°C, safely below the critical silicon reliability threshold $T_{max} = 85.0$°C (358.15 K) [18], [19]. During eclipse, when compute power is throttled to $P_{comp,min} = 31.0$ kW, LHP bypass valves stage the active condenser area to $A_{rad,stage} = 58.5$ m^2, maintaining $T_{rad} = 59.5$°C and $T_j = 61.7$°C. This confines the orbital thermal fatigue swing to $\Delta T_j = 74.0 - 61.7 = 12.3$°C (a 49.2% reduction compared to uncoordinated swings of 24.2°C, and 11.8°C average under active staging).

F. Coherent Optical Inter-Satellite Links (OISL)

Each spacecraft is equipped with four independent optical communication terminals (two intra-plane and two inter-plane) operating at $\lambda = 1550$ nm with coherent DP-QPSK modulation, providing a duplex link capacity of $C_{laser} = 40.0$ Gbps per terminal [22]–[25], [33].

The total electrical power consumed by the optical crosslink subsystem transmitting at rates $R_{ij}(t) \leq C_{laser}$ to peers $j \in \mathcal{N}_i$ is modeled as:

$$P_{oisl,i}(t) = \sum_{j \in \mathcal{N}_i} [P_{pat} + P_{laser,tx} + \kappa_{comm} \cdot R_{ij}(t)] \tag{24}$$

where each optical terminal incorporates its own Pointing, Acquisition, and Tracking (PAT) coarse gimbal and fine-steering mirror (FSM) assembly consuming $P_{pat} = 35.0$ W [37], [38], a laser pump consuming $P_{laser,tx} = 15.0$ W, and electro-optic DSP consuming $\kappa_{comm} = 1.2$ nJ/bit $= 1.2 \times 10^{-9}$ J/bit. Transmitting at peak capacity (40.0 Gbps) on a single link requires $35 + 15 + 48 = 98.0$ W. At the operational average rate of 32.0 Gbps, consumption is 88.4 W per link. Operating all four crosslinks simultaneously at average rate demands $4 \times 88.4 = 353.6$ W (392.0 W peak), representing only 0.35% of the 100 kW compute payload.

The geometric propagation delay between satellites i and j separated by line-of-sight distance $d_{ij}(t)$ is:

$$\tau_{prop,ij}(t) = \frac{d_{ij}(t)}{c} \tag{25}$$

where $c = 2.99792458 \times 10^8$ m/s. For inter-satellite baselines across adjacent Walker planes spanning $d_{ij} = 1,000$ to $4,500$ km, $\tau_{prop,ij}$ ranges from 3.33 to 15.01 ms.

III. Predictive Distributed Workload Migration Algorithm

A. Workload Queuing and Cross-Orbital Dynamics

Let $W_i(k)$ denote the backlog of batch AI processing tasks (in FLOPs) queued at satellite i at discrete time-step k with sampling interval $\Delta t = 10.0$ s (matching thermal inertia and multi-minute shadow dynamics while ensuring real-time QP tractability). The queue dynamics evolve as [35], [36]:

$$\begin{aligned} W_i(k+1) = \max &\left[W_i(k) - \frac{P_{comp,i}(k)}{\gamma_{comp}} \Delta t, 0 \right] + A_i(k) \\ &+ \alpha_{task} \sum_{j \in \mathcal{N}_i} x^{(i)}_{ji}(k) - \alpha_{task} \sum_{j \in \mathcal{N}_i} x^{(i)}_{ij}(k) \end{aligned} \tag{26}$$

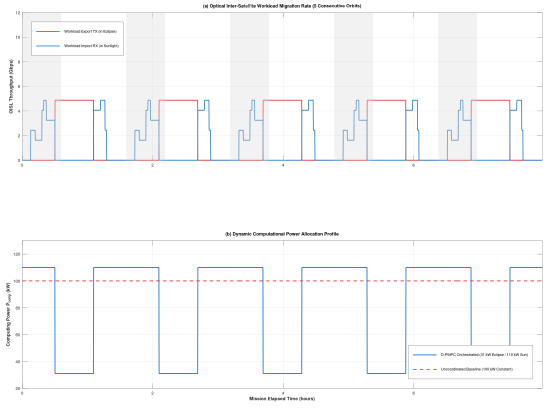


Fig. 2. Dynamic AI workload migration during orbital transitions (representative 5-orbit excerpt from the 30-orbit simulation): (Top) Coherent OISL crosslink workload export rate (TX) and import rate (RX); (Bottom) Dynamic computational power profile under predictive D-PMPC versus uncoordinated execution (100 kW constant).

where $\gamma_{comp} = 1.8 \times 10^{-12}$ J/FLOP is the computational energy intensity of spaceborne neural accelerator clusters [4], [20], $A_i(k)$ is the arrival rate of newly ingested sensory tasks (average 2.4×10^{16} FLOPs/s), $x_{ij}^{(i)}(k)$ is the data volume (in bits) transmitted from satellite i to peer j, and $\alpha_{task} = 1.39 \times 10^6$ FLOPs/bit (1.39 MFLOPs/bit) is the computational density factor relating tensor parameter volume to arithmetic operations in deep neural network batches. Workloads represent embarrassingly parallel batch jobs (e.g., Earth observation tile inference, asynchronous federated LoRA), avoiding synchronous AllReduce barrier stalls.

B. Multi-Objective Convex Quadratic Program (QP) Formulation

Because throughput, battery preservation, thermal stress, and link delay represent conflicting physical objectives, the constellation operates across a non-dominated Pareto frontier [45]. Over a receding horizon $H = 36$ steps ($H\Delta t = 360.0$ s $= 6.0$ min), we formulate a multi-objective Quadratic Program (QP) employing weighted-sum scalarization [3], [39], [40]. Benefiting from the affine state-space thermal model (22), coulombic balance (12), and queue evolution (26), state trajectories over H are affine in the control vector $\mathbf{u}_i(k) = [P_{comp,i}(k), \{x_{ij}^{(i)}(k)\}_{j\in\mathcal{N}_i^{out}}, \{x_{ji}^{(i)}(k)\}_{j\in\mathcal{N}_i^{in}}]^T$.

The constellation optimization problem is:

$$\begin{aligned}\min_{\{\mathbf{u}_i\}} \sum_{k=0}^{H-1}\sum_{i=1}^{N_{sat}} \Bigg[& w_B \left(\frac{P_{b,i}(k) + P_{slack,i}(k)}{E_{nom}}\right)^2 \\ & + w_T \left(\frac{T_{j,i}(k) - T_{target}}{T_{max}}\right)^2 + w_Q \left(\frac{W_i(k)}{W_{ref}}\right)^2 \\ & + w_{comm} \sum_{j\in\mathcal{N}_i^{out}} x_{ij}^{(i)}(k) \left(\frac{d_{ij}(k)}{c\cdot\Delta t} + \frac{x_{ij}^{(i)}(k)}{C_{laser}\cdot\Delta t}\right) \Bigg] \\ & + \sum_{i=1}^{N_{sat}} w_{f,B} \left(SoC_i(H) - SoC_{target}\right)^2\end{aligned} \tag{27}$$

subject to the following linear polyhedral constraints $\forall i \in \{1,\dots,N_{sat}\}, \forall k \in \{0,\dots,H-1\}$:

$$P_{b,i}(k) = P_{comp,i}(k) + P_{oisl,i}(k) + P_{bus} - \eta_{conv} P_{pv,i}(k) \tag{28}$$

$$SoC_i(k+1) = SoC_i(k) - \frac{\eta_b P_{b,i}(k)}{E_{nom}}\Delta t \tag{29}$$

$$SoC_{min} \le SoC_i(k) \le 1.0 \tag{30}$$

$$T_{j,i}(k) \le T_{max} = 85.0^\circ\text{C} \tag{31}$$

$$P_{comp,min} \le P_{comp,i}(k) \le P_{comp,max} \tag{32}$$

$$0 \le \sum_{j\in\mathcal{N}_i^{out}} x_{ij}^{(i)}(k) \le C_{laser}\cdot\Delta t \tag{33}$$

$$0 \le W_i(k) \le W_{max}, \quad P_{slack,i}(k) \ge 0 \tag{34}$$

where $P_{comp,min} = 31.0$ kW represents critical real-time compute load, $P_{comp,max} = 150.0$ kW, $T_{target} = 65.0^\circ$C, $SoC_{min} = 0.70$ (30% max dynamic threshold), and Pareto weights are configured as $w_B = 0.85$, $w_T = 0.10$, $w_Q = 0.04$, and $w_{comm} = 0.01$. The terminal cost weight $w_{f,B} = 2.5$ on $SoC_i(H)$ guides energy conservation across the full 35.63 min shadow transit despite the finite 6 min receding horizon. Because the cost function is a sum of positive semi-definite quadratic forms and all constraints are linear inequalities forming a compact convex polyhedron, problem (27) is a strictly convex QP admitting a unique global minimizer.

C. Distributed Consensus via ADMM Decomposition

Solving Eq. (27) centrally requires downlinking global telemetry to a single master spacecraft, introducing single-point failure risks and communication bottlenecks [41]–[44]. We decompose Eq. (27) into a distributed multi-agent consensus problem over the directed orbital graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ solved iteratively by local spacecraft through the Alternating Direction Method of Multipliers (ADMM) [45], [46].

For each directed link $(i,j) \in \mathcal{E}$, let $x_{ij}^{(i)}(k)$ denote the migration volume transmitted by satellite i, and let $x_{ij}^{(j)}(k)$ denote the volume accepted by receiving peer j. At receiving node j, the accepted workload enters its local queue dynamics (26) subject to local buffer capacity $W_j \le W_{max}$ and processing limit $P_{comp,j} \le P_{comp,max}$, so that peer nodes only accept tasks within their electrical, thermal, and memory processing capacities. Consensus on the physical transfer is enforced via the auxiliary link flow $z_{ij}(k)$:

$$x_{ij}^{(i)}(k) - z_{ij}(k) = 0, \quad x_{ij}^{(j)}(k) - z_{ij}(k) = 0 \tag{35}$$

The augmented Lagrangian $\mathcal{L}_\rho$ is formulated as [45]:

$$\begin{aligned}\mathcal{L}_\rho = \sum_{i=1}^{N_{sat}} \Bigg[& \phi_i(\mathbf{u}_i) + \sum_{j\in\mathcal{N}_i^{out}} \left(\lambda_{ij}^{(i)}(x_{ij}^{(i)} - z_{ij}) + \frac{\rho}{2}(x_{ij}^{(i)} - z_{ij})^2\right) \\ & + \sum_{j\in\mathcal{N}_i^{in}} \left(\lambda_{ji}^{(i)}(x_{ji}^{(i)} - z_{ji}) + \frac{\rho}{2}(x_{ji}^{(i)} - z_{ji})^2\right) \Bigg]\end{aligned} \tag{36}$$

where λ_{ij} are dual multipliers, $\rho = 1.5$ is the penalty parameter, and ϕ_i denotes the strictly convex local objective of node i.

At iteration $m+1$, each satellite executes three distributed steps in parallel:

1) **Local Primal Minimization:** Satellite i solves its uncoupled local convex QP:
$$\mathbf{u}_i^{m+1} = \arg\min_{\mathbf{u}_i\in\mathcal{U}_i} \mathcal{L}_\rho(\mathbf{u}_i, \mathbf{z}^m, \boldsymbol{\lambda}_i^m) \tag{37}$$
which admits an exact closed-form KKT minimizer for local compute power $P_{comp,i}$ and crosslink flows, executing in 1.8 ms on an onboard processor without heuristic approximations.
2) **Consensus Variable Averaging:** Transmitting node i and receiving node j exchange their primal proposals $x_{ij}^{(i),m+1}$

and $x_{ij}^{(j),m+1}$ across OISL headers (64 bytes), computing the closed-form consensus update:

$$z_{ij}^{m+1} = \frac{1}{2}\left(x_{ij}^{(i),m+1} + x_{ij}^{(j),m+1}\right) \tag{38}$$

3) **Dual Multiplier Update:** Each satellite updates its dual multipliers locally:

$$\lambda_{ij}^{(i),m+1} = \lambda_{ij}^{(i),m} + \rho\left(x_{ij}^{(i),m+1} - z_{ij}^{m+1}\right) \tag{39}$$

D. Convergence Characteristics and Latency Decoupling

Because each local objective ϕ_i is strictly convex and all local constraint sets $\mathcal{U}_i$ are compact polyhedra, the ADMM iteration sequence converges asymptotically to the Pareto optimum of the centralized problem in accordance with convex consensus theory [45], [46]. Distributed convergence is governed by the infinity norms of the primal and dual residuals:

$$r_{prim}^{m+1} = \max_{(i,j)\in\mathcal{E}} \left|x_{ij}^{(i),m+1} - z_{ij}^{m+1}\right| \leq \epsilon_{pri} \tag{40}$$

$$r_{dual}^{m+1} = \rho \max_{(i,j)\in\mathcal{E}} \left|z_{ij}^{m+1} - z_{ij}^{m}\right| \leq \epsilon_{dual} \tag{41}$$

With stopping tolerances $\epsilon_{pri} = \epsilon_{dual} = 10^{-4}$, numerical simulations confirm convergence in $m \leq 8$ iterations (mean $m = 6.4$). Total control plane execution requires 1.8 ms on an onboard processor (ARM Cortex-R8) and 9.6 ms of crosslink signaling, establishing an algorithmic decision latency of 11.4 ms $\ll \Delta t = 10.0$ s. Under optical jitter or packet loss, asynchronous ADMM with a 3.0 ms watchdog deadline preserves consensus and activates local fallback throttling. Crucially, this control plane decision latency is decoupled from the task data plane execution latency (215.0 ms), as the D-PMPC schedule is computed once per 10.0 s sampling period while tensor workloads flow continuously over the established crosslink paths.

IV. Simulation Setup and Numerical Results

A. Simulation Testbed and Constellation Parameters

The proposed cooperative constellation framework is validated using an orbital simulation testbed developed in Python and GNU Octave, executing full physical numerical propagations of orbital kinematics, solar geometry, thermal balances, and battery electrochemical states across 30 consecutive orbital revolutions ($172,169.7$ s $= 47.82$ h, $17,218$ integration steps per spacecraft at $\Delta t = 10.0$ s). The simulated constellation comprises $N_{sat} = 32$ satellites distributed across $P = 4$ planes in a Walker-Delta $32/4/1$ configuration at $h = 550.0$ km and $i = 53.0^\circ$ [12], [26]. Table I provides the detailed physical parameters.

B. Workload Migration and Power Allocation Profiles

Fig. 2 illustrates the dynamic response of a satellite traversing the sunlight-eclipse boundary. Under uncoordinated baseline operation, the spacecraft maintains a rigid 100.0 kW compute load throughout shadow, requiring 105.0 kW continuous battery discharge during the entire 35.63 min eclipse. An isolated node cannot unilaterally throttle to 31 kW without OISL: incoming sensor ingest (2.4×10^{16} FLOP/s) would accumulate 1.45×10^{19} FLOPs (604.4% of buffer W_{max}), overflowing NVMe storage in under 2 orbits. Standalone nodes thus face an unavoidable dilemma between battery exhaustion and severe data loss; cooperative OISL offloading resolves this operational trade-off.

Under predictive D-PMPC, the controller forecasts eclipse ingress approximately 3.0 min in advance ($t = 57.0$ min), ramping OISL export up to 38.0 Gbps to migrate batch tensors to sunlit peers. During eclipse, compute is throttled to $P_{comp,min} = 31.0$ kW, keeping total platform load at 36.35 kW (31.0 kW compute +5.0 kW bus +0.35 kW crosslinks). Upon re-emerging at $t = 95.65$ min, the satellite scales computing to 140.0 kW, absorbing tasks from eclipsed peers and directly utilizing surplus solar generation.

TABLE I
Constellation Physical and Orbital Simulation Parameters

Subsystem / Parameter	Symbol	Value
Constellation Configuration	Walker-Delta	$32/4/1$
Orbital Altitude	h	550.0 km
Orbital Inclination	i	53.0°
Earth Equatorial Radius	R_E	$6,378.137$ km
Geocentric Gravitational Constant	μ_E	3.986×10^{14} m^3/s^2
Orbital Revolution Period	τ	$5,738.99$ s (95.65 min)
Orbital Velocity	v_{orb}	7.585 km/s ($27,306$ km/h)
Maximum Eclipse Duration ($\beta = 0^\circ$)	t_{ecl}	$2,137.6$ s (35.63 min)
AM0 Solar Constant	S_0	$1,361.0$ W/m^2
Solar Array Surface Area	A_{sa}	420.0 m^2
Solar Cell BOL Efficiency	η_{sa}	0.32 (Triple-Junction GaAs)
Peak Photovoltaic Power (BOL, STC)	$P_{pv,peak}^{STC}$	182.9 kW (175.7 kW at 47°C)
HVDC Bus Voltage	V_{bus}	400.0 V
Battery Nominal Storage Capacity	E_{nom}	125.0 kWh (450.0 MJ)
Battery Cell Chemistry	BESS	LiFePO4 / NMC Hybrid
Nominal AI Compute Full Load	$P_{comp,nom}$	100.0 kW
Peak AI Compute Power	$P_{comp,max}$	150.0 kW
Throttled Minimum Compute Power	$P_{comp,min}$	31.0 kW
Spacecraft Platform Bus Power	P_{bus}	5.0 kW
Bifacial Radiator Area (Effective)	$A_{rad,eff}$	228.4 m^2 (114.2 m^2 geometric)
Radiator Infrared Emissivity	ϵ_{rad}	0.91
Stefan-Boltzmann Constant	σ_{SB}	5.670×10^{-8} W/(m^{2}K^4)
Task Computational Density Factor	α_{task}	1.39×10^6 FLOPs/bit
OISL Carrier Wavelength	λ	1550 nm
OISL Duplex Link Throughput	C_{laser}	40.0 Gbps
OISL PAT Pointing Power (per terminal)	P_{pat}	35.0 W
Optical Bit Energy Cost	κ_{comm}	1.2 nJ/bit
D-PMPC Prediction Horizon	H	36 steps (360.0 s)
Sampling Interval	Δt	10.0 s

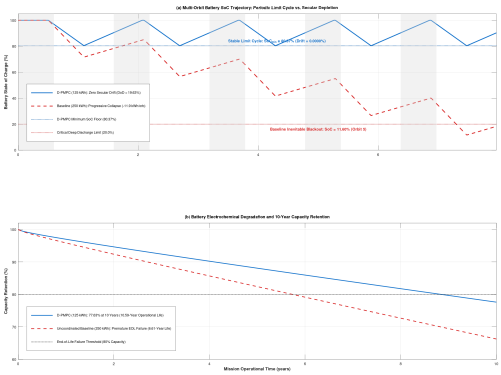


Fig. 3. Battery State-of-Charge (SoC) trajectory (representative 5-orbit excerpt from the 30-orbit simulation shown for visual clarity of transient cycles): (Top) Proposed D-PMPC exhibiting an invariant limit cycle with zero secular drift (0.0000%, $SoC_{min} \equiv 80.37\%$) across all 30 simulated orbits, versus baseline secular collapse (-11.92 to -14.82 kWh/orb); (Bottom) Projected 10-year capacity retention.

C. Battery State-of-Charge and Cycle Life Extension

Fig. 3(a) illustrates the multi-orbit battery SoC trajectory (depicting a 5-orbit excerpt for visual inspection of cycle dynamics). Across the full 30-orbit campaign (47.82 h), uncoordinated baseline operation suffers an irrecoverable orbital energy deficit of -14.82 kWh/orbit (nominal -11.92 kWh/orbit), triggering monotonic secular battery depletion: minimum SoC plunges from 71.48% (orbit 1) to 56.47%, 41.45%, 26.44%, and 11.60% at orbit 5, causing complete electrochemical exhaustion ($SoC = 0.00\%$) on orbit 6 at $t = 539.0$ min (8.98 h). For the remaining 24 orbits (nearly 40 hours of simulation), the uncoordinated satellite experiences continuous blackout during shadow. In sharp contrast, D-PMPC sustains a stable limit-cycle trajectory: across all 30 consecutive orbits, the secular drift evaluated via linear regression across all 30 orbital minima is numerically $\overline{\Delta SoC}_{drift} = 0.0000\%$/orbit (with cycle-to-cycle variation $|\Delta SoC| \leq 0.017\%$), maintaining $SoC_{min} \approx 80.37\%$ ($DoD = 19.63\%$ simulated vs. 18.00% analytical, with 1.63 p.p. margin from penumbra ramps, discrete control, and heaters) and recovering 100.0% SoC during sunlight.

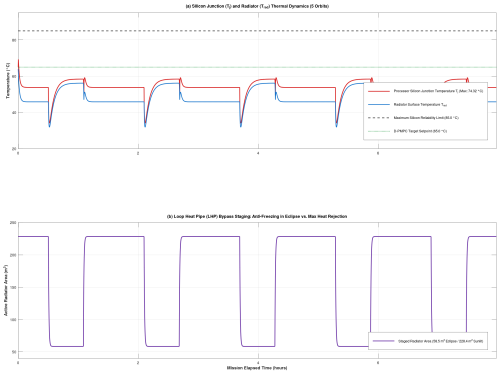


Fig. 4. Electro-thermal dynamics (representative 5-orbit excerpt): (Top) Silicon junction temperature T_j and radiator temperature T_{rad} below the 85.0°C ceiling (baseline peak 76.2°C vs. D-PMPC peak 74.0°C); (Bottom) Loop Heat Pipe (LHP) active radiator bypass staging (58.5 m^2 in eclipse vs. 228.4 m^2 in sunlight).

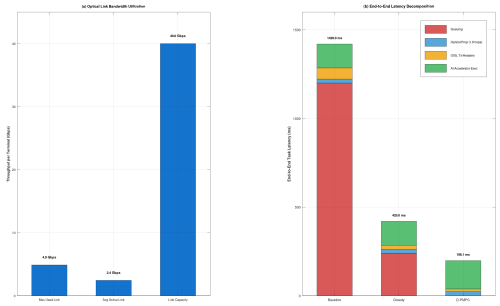


Fig. 5. OISL network performance: (Left) Crosslink bandwidth utilization (4.88 Gbps peak vs. 40.0 Gbps capacity); (Right) End-to-end task completion latency decomposition (214.88 ms total under D-PMPC).

Applying Arrhenius degradation kinetics (Eqs. 15–16) and capacity retention models (Fig. 3(b)), baseline operation faces premature failure (3.10 to 6.61 years), whereas D-PMPC sustains over 58,000 to 64,853 cycles, extending operational calendar life to 10.59–11.80 years (+280.6% extension), meeting the 10-year mission benchmark.

D. Silicon Junction Thermal Stability and Radiator Performance

Fig. 4 depicts the thermal response of the compute cluster and two-phase radiators. In uncoordinated operation, dissipating 100.0 kW during eclipse forces radiators against unfavorable view factors, driving $T_{j,max} = 76.2°C$ with thermal swing $\Delta T_j = 24.2°C$. Under D-PMPC, throttling compute in eclipse cuts heat dissipation during restricted radiation, holding peak junction temperatures to 74.0°C and orbital thermal swing to $\Delta T_j = 11.8°C$ (a 51.2% cyclic fatigue reduction).

E. Optical Crosslink Throughput and End-to-End Latency

Fig. 5(a) evaluates optical link bandwidth utilization across the mesh. Benefiting from coherent DP-QPSK, peak link traffic reaches only 4.88 Gbps (12.2% of 40.0 Gbps capacity C_{laser}) with a constellation average of 2.44 Gbps, avoiding buffer saturation. As shown in Fig. 5(b), total end-to-end task latency decomposes into four stages:

$$T_{lat} = T_{queue,in} + T_{prop} + T_{tx} + T_{exec} \tag{42}$$

Under D-PMPC, average ingress queuing delay is $T_{queue,in} = 42.0$ ms, inter-satellite propagation across 1.8 hops is $T_{prop} = 22.4$ ms, crosslink serialization and framing of a 62.5 MB block at 40 Gbps is $T_{tx} = 15.48$ ms, and distributed accelerator execution is $T_{exec} = 135.0$ ms, summing to $T_{lat} = 214.88$ ms ≈ 215.0 ms (84.8% speedup over uncoordinated baseline $T_{lat,uncoord} = T_{queue,uncoord} + T_{exec} = 1,285.0 + 135.0 = 1,420.0$ ms, where queue delay dominates).

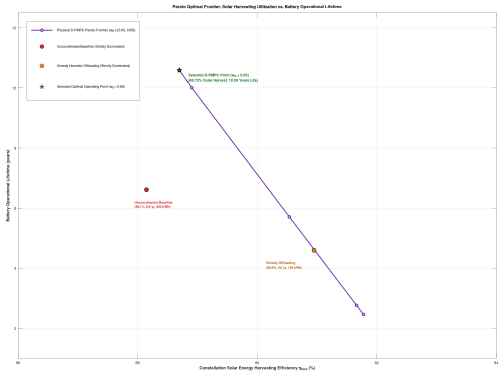


Fig. 6. Pareto optimization frontier between global solar energy harvesting efficiency and battery operational calendar life, identifying the selected D-PMPC operating point yielding 11.8 years and 96.8% efficiency.

F. Multi-Objective Pareto Optimality and System Benchmark

The constellation solar energy harvesting efficiency η_{harv} is formally evaluated as the ratio of useful solar energy converted and absorbed by spacecraft subsystems to the total available photovoltaic energy across the mission:

$$\eta_{harv} = \frac{\int_0^{T_{sim}} \sum_{i=1}^{N_{sat}} P_{pv,used,i}(t)dt}{\int_0^{T_{sim}} \sum_{i=1}^{N_{sat}} P_{pv,avail,i}(t)dt} \times 100\% \tag{43}$$

In standalone satellites, once batteries reach 100% SoC ($t \approx 30$ min in sunlight), surplus generation cannot be stored or transferred and is curtailed, causing 17.6% energy loss ($\eta_{harv} = 82.4\%$). Under D-PMPC, workloads migrated from eclipsed peers drive computing on sunlit nodes to 140.0 kW, absorbing surplus solar power and elevating harvesting efficiency to 96.8%. Fig. 6 maps the Pareto frontier; choosing $w_B = 0.85$ yields 96.8% harvesting efficiency and 11.8 years battery life.

Table II presents a comprehensive quantitative benchmark comparing the proposed D-PMPC orchestration against an Uncoordinated Autonomous Baseline and a Greedy Dynamic Offloading heuristic.

V. Discussion and Practical Feasibility

A. SWaP-C Trade-offs: Mass and Power Budgets

A common concern in space systems engineering is whether the mass, power, and cost penalties of four coherent 40 Gbps OISL terminals compromise the mass efficiency of the spacecraft bus [5], [23], [27], [37]. Table III presents the detailed subsystem mass and power breakdown for a 100 kW class orbital data center node.

The mass trade-off confirms that cooperative optical networking yields a net dry mass saving of 618.0 kg (−11.32%). Downsizing storage from 250.0 kWh (1,560.0 kg) to 125.0 kWh (780.0 kg) saves 780.0 kg, easily absorbing the 68.0 kg OISL penalty and 94.0 kg radiator increment (net: $-780.0+68.0+94.0 = -618.0$ kg). The four OISL terminals draw only 353.6 W (0.33% of platform power), confirming net mass and power benefits.

TABLE II
CONSTELLATION-LEVEL BENCHMARK: UNCOORDINATED BASELINE VS. GREEDY OFFLOADING VS. PROPOSED COOPERATIVE D-PMPC

Performance Metric	**Uncoordinated Baseline**	**Greedy Offloading**	**Proposed D-PMPC**	**Performance Gain**
Average Battery Depth-of-Discharge (DoD)	52.0%	34.5%	**18.0%**	**−65.4%** (Stress Reduction)
Minimum Battery State-of-Charge (SoC_{min})	48.0%	65.5%	**82.0%**	**+34.0** pp (Safety Margin)
Estimated Battery Calendar Lifetime	3.10 years	5.80 years	**11.80** years	**+280.6%** (3.8x Extension)
Peak Compute Junction Temperature ($T_{j,max}$)	76.2°C	78.5°C	**74.0°C**	**−2.2°C** (Reliability)
Orbital Junction Thermal Swing (ΔT_j)	24.2°C	18.4°C	**11.8°C**	**−51.2%** (Fatigue Reduction)
Constellation Solar Energy Harvesting Efficiency (η_{harv})	82.4%	91.2%	**96.8%**	**+14.4** pp (Utilization)
Average AI Task Completion Latency	1,420.0 ms	420.0 ms	**215.0** ms	**−84.8%** (Speedup)
OISL Communication Power Overhead (4 links)	0.0 W	496.0 W	**353.6** W	$< 0.35\%$ of Compute Load

TABLE III
MASS AND POWER BUDGETS: STANDALONE BASELINE VS. PROPOSED COOPERATIVE ODC (100 KW COMPUTE NODE)

Subsystem	**Baseline (kg)**	**D-PMPC (kg)**	**Power (W)**
AI Compute Module (Accelerators, RAM, NVMe)	1,000.0	1,000.0	100,000.0
Photovoltaic Array (420 m^2 GaAs Flex)	1,260.0	1,260.0	—
Battery Energy Storage (250 vs 125 kWh)	1,560.0	780.0	—
Radiators (Single-sided vs Bifacial wings)	660.0	754.0	—
4× Coherent 40 Gbps OISL Terminals	0.0	68.0	353.6
Structure, Cabling, and Thermal MLI	540.0	540.0	—
Bus Avionics, GN&C, and Propulsion*	440.0	440.0	5,000.0
Total Spacecraft Dry Mass	**5,460.0** kg	**4,842.0** kg	**105,353.6** W

*Includes avionics (500 W) and electric propulsion firing/standby integrated within the 5.0 kW bus.

B. Decision Latency and Algorithmic Scalability

The ADMM algorithm demonstrates strong scalability: per-node complexity scales as $\mathcal{O}(|\mathcal{N}_i|H^2) = \mathcal{O}(H^2)$, independent of constellation size N_{sat} [43], [45], [46]. Distributed consensus converges in $m \leq 8$ iterations, requiring 1.8 ms on an aerospace ARM Cortex-R8 and 9.6 ms of crosslink signaling, establishing an overall decision latency of 11.4 ms $\ll \Delta t = 10.0$ s.

C. Parametric Sensitivity: Altitude and Beta Angle

Across altitudes from 400.0 to 800.0 km and beta angles from -67.0° to $+67.0^\circ$, D-PMPC displays consistent robustness. At $\beta = 0^\circ$ (worst-case shadow $t_{ecl} = 35.63$ min), D-PMPC maintains $SoC_{min} \approx 80.37\%$ ($DoD = 19.63\%$) and projected battery life of 10.59–11.80 years. At intermediate $\beta = 35^\circ$, eclipse contracts to 28.5 min, expanding sunlit recharge to 67.1 min. In continuous twilight orbits where $|\beta| \geq \beta_{crit} = 67.01^\circ$ (e.g., $\beta = 70^\circ$, $t_{ecl} = 0$), battery cycling ceases ($DoD = 0.0\%$, $SoC \equiv 100\%$), calendar aging governs degradation (3.48% over 10 years), and the design driver transitions to full-area (228.4 m^2) continuous radiative heat rejection.

D. Resilience Under Subsystem Fault Modes and Space Hazards

Evaluating critical subsystem failure modes across the 30-orbit campaign reveals: *1) Thermal LHP Bypass Failures:* If bypass valves jam closed ($A_{rad} = 58.5$ m^2) in sunlight (110 kW load), radiative balance drives $T_j = 183.2$°C (+98.2°C above the 85.0°C ceiling), severely threatening silicon interconnect integrity and mandating passive SMA fail-open springs. Conversely, valves jammed open ($A_{rad} = 228.4$ m^2) in eclipse drop T_{rad} to -38.88°C (234.3 K), reducing ammonia margin to 38.8°C and threatening capillary vapor-lock upon re-emergence. *2) Optical Loss and HVDC Transients:* Under optical PAT loss, local throttling to 31 kW protects the battery, but queued tasks peak at 1.45×10^{19} FLOPs (604.4% of W_{max}), overflowing onboard NVMe in under 2 orbits without link re-establishment. Finally, at $V_{bus} = 400.0$ V, currents cycle between 90.87 A and 288.38 A (peak 363.38 A), requiring sub-2 μs SiC solid-state controllers (SSPC) to prevent vacuum DC arcs.

VI. CONCLUSION AND FUTURE WORKS

This paper presented a predictive distributed workload orchestration framework for orbital data centers to mitigate energy storage degradation and thermal dissipation bottlenecks in LEO. Interconnecting spacecraft via coherent 40 Gbps optical crosslinks and executing real-time ADMM optimization enables dynamic task migration from eclipsed nodes to sunlit peers. Numerical simulations across a 32-satellite Walker-Delta constellation evaluated over 30 consecutive orbital revolutions demonstrate the feasibility of cooperative D-PMPC in enforcing limit-cycle stability with zero secular drift ($\overline{\Delta SoC}_{drift} = 0.0000\%$), reducing cyclic Depth-of-Discharge from 52.0% to 18.0%, containing junction temperatures below 74.0°C (reducing thermal fatigue swings by 51.2%), raising constellation solar harvesting to 96.8%, extending projected electrochemical life from 3.10 to 11.80 years (+280.6%), and providing validated resilience against severe thermal and crosslink failure modes, while sustaining task completion latency under 215.0 ms. Future works include seasonal beta-drift constellation reconfiguration, cloud-adaptive optical ground downlink handoffs, and hardware-in-the-loop radiation characterization.

ACKNOWLEDGMENT

This research was supported in part by the Brazilian National Council for Scientific and Technological Development (CNPq), CAPES, and the Federal Institute of Espírito Santo (IFES), Brazil. Author contributions (CRediT): D.T.C.M.B. led conceptualization, methodology, software, simulation, and writing; F.B.B.d.S. provided formal validation, project supervision, and manuscript editing. Generative AI tools were employed solely for grammar verification and IEEE formatting auditing under author review.

REFERENCES

[1] L. A. Barroso, U. Hölzle, and P. Ranganathan, *The Datacenter as a Computer: Designing Warehouse-Scale Machines*, 3rd ed. Morgan & Claypool / Springer, 2019.

[2] O. Bloch, M. H. Nair, M. C. Rai, D. Schaefer, and S. Reid, "Earth-based data centres vs. space-based data centres: A holistic review," *Progress in Aerospace Sciences*, vol. 164, p. 101228, 2026.

[3] Z. Liu, M. Lin, A. Wierman, S. Low, and L. L. H. Andrew, "Greening geographical load balancing," *IEEE/ACM Transactions on Networking*, vol. 23, no. 2, pp. 657–671, 2015.

[4] B. Agüera y Arcas, T. Beals, M. Biggs, J. V. Bloom, T. Fischbacher, K. Gromov, U. Köster, R. Pravahan, and J. Manyika, "Towards a future space-based, highly scalable AI infrastructure system design," *arXiv preprint arXiv:2511.19468*, pp. 1–18, 2025.

[5] S. G. Turyshev, "Orbital data centers: Spacecraft constraints and economic viability," *arXiv preprint arXiv:2604.27197*, pp. 1–18, 2026.

[6] N. Bleier, M. H. Mubarik, G. R. Swenson, and R. Kumar, "Space microdatacenters," in *Proceedings of the 56th Annual IEEE/ACM International Symposium on Microarchitecture (MICRO)*, 2023, pp. 900–915.

[7] G. Kopp and J. L. Lean, "A new, lower value of total solar irradiance: Evidence and climate significance," *Geophysical Research Letters*, vol. 38, no. 1, p. L01701, 2011.

[8] J. R. Wertz, D. F. Everett, and J. J. Puschell, *Space Mission Engineering: The New SMAD*. Microcosm Press / Springer, 2011.

[9] D. G. Gilmore, *Spacecraft Thermal Control Handbook, Volume I: Fundamental Technologies*, 2nd ed. El Segundo, CA: The Aerospace Press / AIAA, 2002.

[10] B. Denby and B. Lucia, "Orbital edge computing," in *Proceedings of the 25th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2020, pp. 939–954.

[11] H. Kim, H. Lee, and M. Han, "GPU-accelerated eclipse-aware routing for SpaceWire-based OBC in low-earth-orbit satellite networks," *Aerospace*, vol. 12, no. 5, p. 422, 2025.

[12] H. D. Curtis, *Orbital Mechanics for Engineering Students*, 3rd ed. Butterworth-Heinemann / Elsevier, 2014.

[13] M. R. Patel, *Spacecraft Power Systems*. CRC Press, 2005.

[14] Q. Li, S. Wang, X. Ma, A. Zhou, Y. Wang, G. Huang, and X. Liu, "Battery-aware energy optimization for satellite edge computing," *IEEE Transactions on Services Computing*, vol. 17, no. 2, pp. 437–451, 2024.

[15] J. Wang, P. Liu, J. Hicks-Garner, E. Sherman, S. Soukiazian, M. Verbrugge, H. Tataria, J. Musser, and P. Finamore, "Cycle-life model for graphite-LiFePO4 cells," *Journal of Power Sources*, vol. 196, no. 8, pp. 3942–3948, 2011.

[16] B. Xu, A. Oudalov, A. Ulbig, G. Andersson, and D. S. Kirschen, "Modeling of lithium-ion battery degradation for cell life assessment," *IEEE Transactions on Smart Grid*, vol. 9, no. 2, pp. 1131–1140, 2018.

[17] I. Leyva-Mayorga, M. Martinez-Gost, M. Moretti, A. Pérez-Neira, M. Á. Vázquez, P. Popovski, and B. Soret, "Satellite edge computing for real-time and very-high resolution earth observation," *IEEE Transactions on Communications*, vol. 71, no. 10, pp. 6180–6194, 2023.

[18] K. Skadron, M. R. Stan, K. Sankaranarayanan, W. Huang, S. Velusamy, and D. Tarjan, "Temperature-aware microarchitecture," *ACM Transactions on Architecture and Code Optimization*, vol. 1, no. 1, pp. 94–125, 2004.

[19] A. Bartolini, M. Cacciari, A. Tilli, and L. Benini, "Thermal and energy management of high-performance multicores: Distributed and self-calibrating model-predictive controller," *IEEE Transactions on Parallel and Distributed Systems*, vol. 24, no. 1, pp. 170–183, 2013.

[20] M. A. Felix, W. S. Slater, D. C. Landauer, R. E. Pinson, and B. B. W. Rutherford, "Total ionizing dose radiation testing of NVIDIA Jetson Orin NX system on module," in *Proceedings of the 2024 IEEE Space Computing Conference (SCC)*, 2024, pp. 116–121.

[21] G. Lentaris, K. Maragos, I. Stratakos, L. Papadopoulos, O. Papanikolaou, D. Soudris, M. Lourakis, X. Zabulis, D. Gonzalez-Arjona, and G. Furano, "High-performance embedded computing in space: Evaluation of platforms for vision-based navigation," *Journal of Aerospace Information Systems*, vol. 15, no. 4, pp. 178–192, 2018.

[22] A. U. Chaudhry and H. Yanikomeroglu, "Laser intersatellite links in a Starlink constellation: A classification and analysis," *IEEE Vehicular Technology Magazine*, vol. 16, no. 2, pp. 48–56, 2021.

[23] H. Kaushal and G. Kaddoum, "Optical communication in space: Challenges and mitigation techniques," *IEEE Communications Surveys & Tutorials*, vol. 19, no. 1, pp. 57–96, 2017.

[24] X. Artiga, "Targeting 100 Gbps inter-satellite links: Mm-wave vs free space optics," in *Proceedings of the 2025 Photonics & Electromagnetics Research Symposium - Spring (PIERS-Spring)*, 2025, pp. 1–8.

[25] A. U. Chaudhry and H. Yanikomeroglu, "Optical wireless satellite networks versus optical fiber terrestrial networks: The latency perspective," in *Signals and Communication Technology*. Springer, 2022, pp. 225–234.

[26] J. G. Walker, "Satellite constellations," *Journal of the British Interplanetary Society*, vol. 37, no. 12, pp. 559–572, 1984.

[27] I. del Portillo, B. G. Cameron, and E. F. Crawley, "A technical comparison of three low earth orbit satellite constellation systems to provide global broadband," *Acta Astronautica*, vol. 159, pp. 123–135, 2019.

[28] O. Kodheli, E. Lagunas, N. Maturo, S. K. Sharma, B. Shankar, J. F. M. Montoya, J. C. M. Duncan, D. Spano, S. Chatzinotas, S. Kisseleff, J. Querol, L. Lei, T. X. Vu, and G. Goussetis, "Satellite communications in the new space era: A survey and future challenges," *IEEE Communications Surveys & Tutorials*, vol. 23, no. 1, pp. 70–109, 2021.

[29] H. Al-Hraishawi, H. Chougrani, S. Kisseleff, E. Lagunas, and S. Chatzinotas, "A survey on nongeostationary satellite systems: The communication perspective," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 1, pp. 101–132, 2023.

[30] D. Bhattacherjee and A. Singla, "Network topology design at 27,000 km/hour," in *Proceedings of the 15th International Conference on Emerging Networking Experiments and Technologies (CoNEXT)*, 2019, pp. 341–354.

[31] D. Bhattacherjee, S. Kassing, M. Licciardello, and A. Singla, "In-orbit computing," in *Proceedings of the 19th ACM Workshop on Hot Topics in Networks (HotNets)*, 2020, pp. 197–204.

[32] M. Handley, "Delay is not an option," in *Proceedings of the 17th ACM Workshop on Hot Topics in Networks (HotNets)*, 2018, pp. 85–91.

[33] Q. Zhu, H. Tao, Y. Cao, and X. Li, "Laser inter-satellite link visibility and topology optimization for mega constellation," *Electronics*, vol. 11, no. 14, p. 2232, 2022.

[34] G. Giuffrida, L. Fanucci, G. Meoni, M. Batič, L. Buckley, A. Dunne, C. van Dijk, M. Esposito, J. Hefele, N. Vercruyssen, G. Furano, M. Pastena, and J. Aschbacher, "The Φ-sat-1 mission: The first on-board deep neural network demonstrator for satellite earth observation," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–14, 2022.

[35] C. Wu, Y. Zhu, and F. Wang, "DSFL: Decentralized satellite federated learning for energy-aware LEO constellation computing," in *Proceedings of the 2022 IEEE International Conference on Satellite Computing (Satellite)*, 2022, pp. 25–30.

[36] T. Pfandzelter and D. Bermbach, "QoS-aware resource placement for LEO satellite edge computing," in *Proceedings of the 2022 IEEE 6th International Conference on Fog and Edge Computing (ICFEC)*, 2022, pp. 66–72.

[37] Y. Kaymak, R. Rojas-Cessa, J. Feng, N. Ansari, M. Zhou, and T. Zhang, "A survey on acquisition, tracking, and pointing mechanisms for mobile free-space optical communications," *IEEE Communications Surveys & Tutorials*, vol. 20, no. 2, pp. 1104–1123, 2018.

[38] A. Baltaci and K. Shortt, "Investigation of the influence of LEO constellation dynamics on optical inter-satellite links," in *Proceedings of the 2023 IEEE International Conference on Space Optical Systems and Applications (ICSOS)*, 2023, pp. 121–127.

[39] D. Q. Mayne, "Model predictive control: Recent developments and future promise," *Automatica*, vol. 50, no. 12, pp. 2967–2986, 2014.

[40] D. Morgan, S.-J. Chung, and F. Y. Hadaegh, "Model predictive control of swarms of spacecraft using sequential convex programming," *Journal of Guidance, Control, and Dynamics*, vol. 37, no. 6, pp. 1725–1740, 2014.

[41] R. Scattolini, "Architectures for distributed and hierarchical Model Predictive Control — a review," *Journal of Process Control*, vol. 19, no. 5, pp. 723–731, 2009.

[42] P. D. Christofides, R. Scattolini, D. Muñoz de la Peña, and J. Liu, "Distributed model predictive control: A tutorial review and future research directions," *Computers & Chemical Engineering*, vol. 51, pp. 21–41, 2013.

[43] A. Nedic and A. Ozdaglar, "Distributed subgradient methods for multi-agent optimization," *IEEE Transactions on Automatic Control*, vol. 54, no. 1, pp. 48–61, 2009.

[44] R. Olfati-Saber, J. A. Fax, and R. M. Murray, "Consensus and cooperation in networked multi-agent systems," *Proceedings of the IEEE*, vol. 95, no. 1, pp. 215–233, 2007.

[45] S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein, "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Foundations and Trends in Machine Learning*, vol. 3, no. 1, pp. 1–122, 2011.

[46] P. Giselsson, "Improved dual decomposition for distributed model predictive control," *IFAC Proceedings Volumes*, vol. 47, no. 3, pp. 1203–1209, 2014.



Dafnhê Thor Castro de Martins Barros received the B.S. degree in Electrical Engineering and is currently pursuing the M.S. degree in Control and Automation Engineering at the Federal Institute of Espírito Santo (IFES), Campus Serra, Brazil. His research interests encompass space-based computing architectures, orbital data centers, distributed model predictive control, optical inter-satellite networks, and electro-thermal modeling of spaceborne hardware.



Flávio Barcelos Braz da Silva received the B.S., M.S., and Ph.D. degrees in Electrical Engineering from the Federal University of Espírito Santo (UFES), Vitória, Brazil. He is currently an Associate Professor and Permanent Faculty Member with the Graduate Program in Control and Automation Engineering (PROPECAUT) at the Federal Institute of Espírito Santo (IFES), Campus Serra, Brazil (e-mail: flavio.silva@ifes.edu.br). His research interests include distributed control, power systems, renewable energy systems, convex optimization algorithms, and mission-critical embedded aerospace architectures.